Optimizing waste recycling points in Eindhoven

Currently, in 2026, Eindhoven has two waste recycling points (WRPs / “milieustraten”): Acht and Lodewijkstraat. In 2012, however, there were three waste recycling points in Eindhoven: Esperheide, Lodewijkstraat, and Gabriël Metsulaan. In this assignment we focus on the WRP in the Lodewijkstraat. We go back in time to 2012, when a team of students collected very detailed customer data in this WRP. Your task is to model the waste recycling point using a queueing model and to write a discrete event simulation to get more insight in the most relevant performance characteristics of this WRP and identify the bottlenecks.

Waste recycling is free for residents of Eindhoven, but they have to identify themselves by showing their city pass to the operator at the entrance. Most residents arrive by car or a small van, which form a queue in front of the entrance. The design of the Lodewijkstraat WRP is not optimal: if the queue is longer than three or four cars, the remaining vehicles have to wait on the road. Obviously, this is an undesirable situation and the municipality of Eindhoven would like to improve this situation.

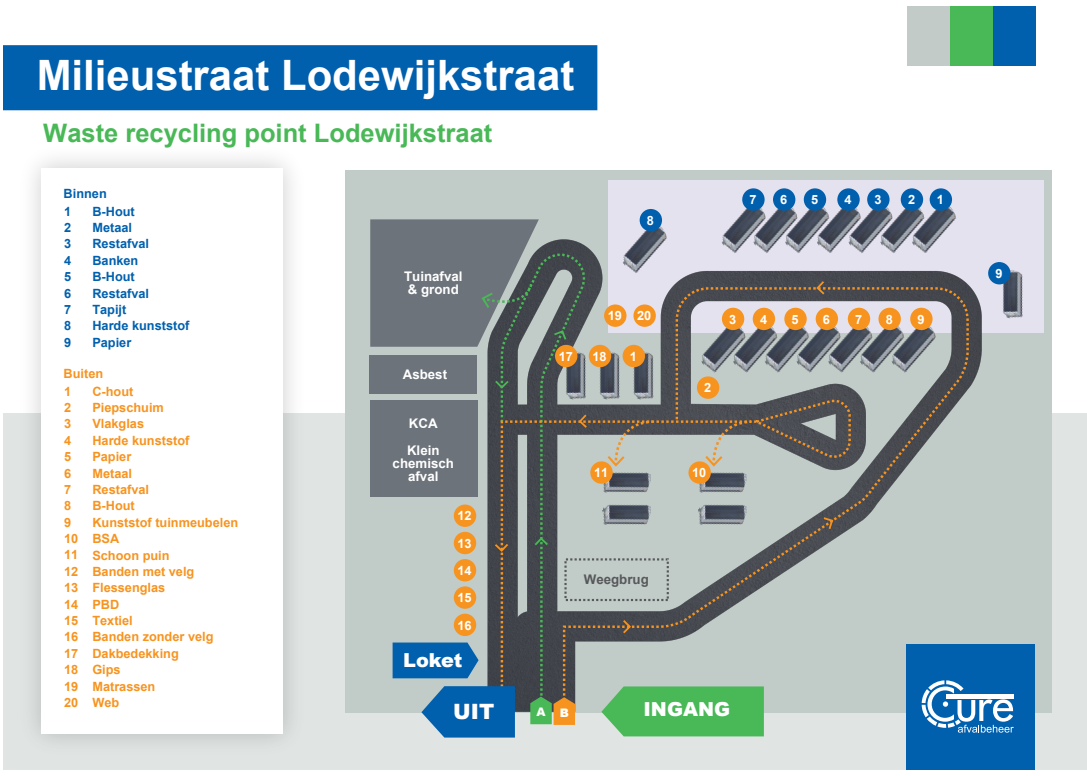

Figure 1: Current layout of the garbage dump at the Lodewijkstraat (source: Cure Afvalbeheer).

Layout of the WRP

For simplicity we will focus on the vast majority of customers arriving by car or van. Most of them bring more than one type of waste, meaning that they will park their vehicle at multiple different locations inside the WRP and dump their waste by foot, before exiting the WRP. In

general, big vehicles (vans or cars with a small trailer) occupy two parking spots, whereas regular cars occupy one spot. The Lodewijkstraat WRP can be divided into six areas (see Figure 2):

- The entrance (or gate). This is where customers show their city pass, which is checked by the WRP employee before granting access. Only one customer can be checked at a time.
- The Hall. This is a typical feature of the Lodewijkstraat WRP. Is a separate queueing system within the WRP, with its own queue in front of it. Inside the hall, there are several containers: six are shown in Figure 2, but as you can see in the official picture (Figure 1), there are actually nine containers ($1-9$). Most importantly, there are 12 parking places inside the hall. Recall, though, that a large car or van will occupy two spaces.
- Overflow. These containers ($1-9$ and $17-20$) are located immediately outside the hall and are intended for small cars only. Many of these contains contain the same garbage types as containers inside the hall. There are 10 parking spaces in this area.
- Debris Clean and Debris Dirty (DcDd). These containers ($10-11$) can only be visited after passing the hall and the overflow areas and contain 7 parking spaces (so max 3 large cars).
- Green ("Tuinafval & grond"). This area is for a separate customer type with its own routing only. It has 5 parking spaces.
- Rest (Asbestos, KCA, containers $12-16$). Can be visited by all customers types. Also has 5 parking spaces.

Inside each of these areas, customers will park their vehicle only once and dump their waste by foot. Customers who need to visit multiple areas, will park exactly once in each area.

Routing and Blocking

In order to prevent (too much) blocking inside the WRP, there are strict rules regarding the routing of customers. There are two customers types: with and without green waste. Stations can only be visited in this order for each of these customer types (see also Figure 1):

A: Entrance $\longrightarrow$ Green $\longrightarrow$ Rest.
B: Entrance $\longrightarrow$ Hall $\longrightarrow$ Overflow $\longrightarrow$ DcDd $\longrightarrow$ Rest.

Customers are not allowed to have a combination of garbage types "Green" and any of the garbage types that require going to the hall, overflow, or debris (clean or dirty) areas. The operator at the entrance checks the city pass and the garbage types of the customer. Then they check if the customer is allowed access to the WRP. If there are no parking places available at the first station a customer needs to visit, the customer is *blocked* and remains at the gate until a parking place becomes available (or two places in case of a large vehicle). Obviously, this means that the customers behind them are also blocked and a queue forms before the entrance.

For Type B customers, there is one more place where a queue can be formed besides the entrance: in front of the hall. Type B customers will *always* first travel through the hall, even if they don't have to dump any garbage inside the hall. The routing, blocking, and queueing rules for these customers are as follows:

Figure 2: A schematic representation of the garbage dump at the Lodewijkstraat.

- If there is a queue in front of the hall of length that would be come longer that the length of 14 small cars (or 7 large cars) by allowing a new car to join this queue, this car is blocked at the entrance. So a small car would be blocked if the current queue length is 14 and a large car would be blocked already at a queue length of 13 cars.
- When there is a blocked car in front of the entrance, all cars behind it are also blocked. This holds for Type A and B cars, despite the fact that Figure 1 seems to suggest otherwise.
- Vehicles visit *either* the hall, *or* the overflow – never both. Of course, some vehicles will visit neither of these two stations.
- The overflow is only accessible to *small* cars; large cars are never allowed in the overflow.
- Large cars will always go to the hall instead of overflow.
- Small cars *prefer* to go to overflow, but if the overflow parking places are all occupied, then (and only then) small cars can go to the hall.
- Type B vehicles that do not need to visit will still need to drive through the hall. This means that they will be blocked too if there is a queue in front of the hall.

Please note the following simplifying assumption that follows from these rules:

We do not really distinguish between hall and overflow when it comes to garbage types. Any small vehicle that wants to visit the overflow station will still be able to dump all their garbage in the hall if the overflow is full.

Another remark concerning blocking inside the WRP:

- A car can only move to the next station if there is a sufficiently large available parking spot there.
- , As long as there is no available parking spot, the car stays in its place, occupying the current parking spot

Example: Suppose that a large car is in the hall and just finished service. It now wants to go to DcDd. However, there is only 1 available (small) parking spot at DcDd. This implies that the car has to wait *inside* the hall for another car to leave at DcDd. While waiting in the hall, it keeps that parking spot occupied and prevents other cars from taking that spot.

Available data

There is one data set available. This data has been cleaned and is immediately ready to use in your simulation. It contains the following sheets:

- Arrival and service processes: information about opening hours, arrival rates throughout the day, fraction big cars, service times of each of the stations, number of parking spaces in each of the areas, maximum queue length in front of the hall.
- Customer classes and routing: Based on a cluster analysis of real data, multiple customer groups have been identified based on the combination of their garbage types. For each of these groups, data is available about which percentage of customers belongs to this group, and how the probabilistic routing takes place. We will illustrate this by the following example for the first customer class, called "Hall, Debris, Green":

Customer class:	Hall, Debris, Green						
Percentage of total:	12%						
Routing matrix				**To**			
			Hall/Overflow	**Debris (clean + dirty)**	**Green**	**Rest**	**Exit**
		Entrance	100%	0%	0%	0%	0%
		Hall/Overflow		13%	31%	0%	56%
	From	**Debris (clean + dirty)**			0%	0%	100%
		Green				0%	100%
		Rest					100%

We conclude the following about this customer class:

  - 12% of all customers belong to this class.
  - These customers only visit Hall/Overflow, Debris, and/or Green.
  - In fact, *all* of these customers visit the hall (100%). After visiting the hall, 13% proceeds to Debris, 31% proceeds to Green, and 56% leaves the WRP directly.
  - All customers that visit either Debris or Green will leave the WRP immediately thereafter. This implies that none of these customers will visit both Debris *and* Green.

Please note that this is old data from 2012 when the routing rules were more lenient. As you might have observed from the above example, in those days customers were allowed to visit both the Hall *and* Green. This is no longer allowed, so in your simulation you will (somehow) have to make sure that you avoid this situation, while still keeping your input data as realistic as possible, based on the provided data from 2012.

Main questions

The municipality of Eindhoven has three main questions:

- Can you simulate the waiting times and queue lengths of the WRP in its current state?
- Eindhoven was planning to close the Esperheide WRP, which would lead up to 20% more people are going to use the Lodewijkstraat WRP. Can you quantify how the WRP will perform with 20% more people than currently visit the WRP (according to the data)?
- Can you suggest improvements that significantly improve this performance? You can think of: shortening service times, adding parking spots, creating different stations, changing the routing.

Your task is to develop a discrete-event simulation that simulates the WRP based on the data that has been provided. You should deliver a well-written report with clear, understandable graphs and tables that support your advice to the municipality of Eindhoven. Each group should hand in a separate PDF file of the report and a zip file containing the source code of their simulation programs. More detailed guidelines (the rubric) and the deadline can be found in Canvas.

Important: Although you are allowed to use AI tools to assist you with this assignment, we stress (again) that you should always honestly cite your sources including an indication of where and how you used AI tools. There are two other restrictions concerning your programming code:

- You are not allowed to use any packages that have not been used during the lectures. If you would like to deviate from this rule, because you really think that another package might be useful for your particular implementation, please contact one of the responsible lecturers to ask for permission.
- Your code should really be based on the sample code provided during the lectures. In particular, classes like the future event set, events, customers, simulation results, and the structure of the main simulation should have a similar setup. The reason why we demand this, is because it really makes it easier for us to assess the correctness of your code, which is needed for the assessment of this assignment. We realise that there are different and probably more elegant ways to implement these types of discrete-event simulations, but if the code looks entirely different, it is much too time consuming for us to grade it. Your code should be easy to understand and well annotated and documented.